

Vibe Commerce - Full Stack E-Commerce Cart Application

A modern, full-stack e-commerce shopping cart application built with React, Node.js, Express, and MongoDB. Features a complete shopping experience with product browsing, cart management, and mock checkout process.

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Core Features

>>Product Catalog: Browse through 8+ mock products with images, prices, and descriptions

>>Shopping Cart: Add, remove, and update product quantities in real-time

>>Cart Persistence: Session-based cart that persists across browser refreshes

>>Checkout Process: Complete mock checkout with customer information form

>>Order Confirmation: Detailed receipt with order summary and customer details

>>Responsive Design: Mobile-first design that works on all devices

Advanced Features

>>RESTful API: Clean, well-structured API endpoints

>>Error Handling: Comprehensive error handling on both frontend and backend

>>Loading States: Smooth loading indicators for better UX

>>Form Validation: Client and server-side form validation

>>Modern React Patterns: Context API, custom hooks, and functional components

>>Database Integration: MongoDB with Mongoose ODM

Tech Stack

Frontend

>>React 18 - Modern React with functional components and hooks

>>Context API - State management for cart and application state

>>Axios - HTTP client for API communication

>>CSS3 - Custom responsive styling with Flexbox and Grid

>>Local Storage - Session persistence

Backend

>>Node.js - Runtime environment

>>Express.js - Web framework

>>MongoDB - NoSQL database

>>Mongoose - MongoDB object modeling

>>CORS - Cross-origin resource sharing

>>UUID - Session ID generation

Development Tools

>Nodemon - Auto-restart server during development

>React Scripts - Create React App build tools

Project Structure

text

vibe-commerce/

```
├── backend/
│   ├── config/
│   │   └── database.js    # MongoDB connection configuration
│   ├── models/
│   │   ├── Product.js    # Product schema and model
│   │   └── Cart.js       # Cart schema and model
│   ├── routes/
│   │   ├── products.js   # Product-related endpoints
│   │   ├── cart.js       # Cart management endpoints
│   │   └── checkout.js   # Checkout processing endpoint
│   ├── middleware/
│   │   └── errorHandler.js # Global error handling middleware
│   ├── scripts/
│   │   └── seed.js       # Database seeding script
│   ├── package.json
│   └── server.js         # Express server entry point
├── frontend/
└── public/
```

```
| | └─ index.html      # HTML template
|   └─ src/
|   |   └─ components/
|   |   |   └─ ProductList.js # Product grid component
|   |   |   └─ Cart.js       # Shopping cart component
|   |   |   └─ Checkout.js   # Checkout form component
|   |   |   └─ Receipt.js    # Order confirmation component
|   |   └─ context/
|   |       └─ CartContext.js # Cart state management
|   └─ services/
|       └─ api.js           # API service layer
|       └─ styles/
|           └─ App.css      # Global styles
|           └─ App.js       # Main application component
|           └─ index.js     # React application entry point
└─ package.json
└─ README.md
```

Installation

Prerequisites

Node.js (v14 or higher) - [Download here](#)

MongoDB (local installation or MongoDB Atlas account)

Git - [Download here](#)

Step 1: Clone and Setup

bash

Clone the repository or create the project structure

```
mkdir vibe-commerce
```

```
cd vibe-commerce
```

Step 2: Backend Setup

```
bash
```

Create and navigate to backend directory

```
mkdir backend
```

```
cd backend
```

Initialize package.json (copy from the code above)

Then install dependencies

```
npm install
```

Step 3: Frontend Setup

```
bash
```

Navigate to project root and create React app

```
cd ..
```

```
npx create-react-app frontend
```

```
cd frontend
```

Install additional dependencies

```
npm install axios
```

Replace default files with the provided code

 Running the Application

Option A: Using Local MongoDB

Start MongoDB Service:

```
bash
```

Windows (Run as Administrator)

```
net start MongoDB
```

Mac (using Homebrew)

```
brew services start mongodb/brew/mongodb-community
```

Linux

```
sudo systemctl start mongod
```

Seed the Database:

```
bash
```

```
cd backend
```

```
npm run seed
```

Start Backend Server:

```
bash
```

```
npm run dev
```

Backend will run on <http://localhost:5000>

Start Frontend Development Server:

```
bash
```

```
cd ../frontend
```

```
npm start
```

Frontend will run on <http://localhost:3000>

I AM USING THIS BEACUSE I DONT HAVE THE PAYED VERSION DO I AM USING THE
Mongodb Atlas

Option B: Using MongoDB Atlas (Cloud - Recommended)

Create MongoDB Atlas Account:

- >Go to MongoDB Atlas
- >Create a free cluster
- >Get your connection string
- >Create Environment File:

```
bash
```

```
cd backend
```

```
touch .env
```

Add to .env file:

```
env
```

```
MONGODB_URI=mongodb+srv://username:password@cluster0.xxxxx.mongodb.net/vi  
becommerce?retryWrites=true&w=majority
```

```
PORT=5000
```

```
NODE_ENV=development
```

🚀 API Documentation

Base URL

text

```
http://localhost:5000/api
```

Products Endpoints

Method	Endpoint	Description	Body
--------	----------	-------------	------

GET	/products	Get all products	-
-----	-----------	------------------	---

GET	/products/:id	Get single product	-
-----	---------------	--------------------	---

Cart Endpoints

Method	Endpoint	Description	Body
--------	----------	-------------	------

GET	/cart	Get cart with total	-
-----	-------	---------------------	---

POST	/cart	Add item to cart	{ productId, quantity }
------	-------	------------------	-------------------------

DELETE	/cart/:id	Remove item from cart	-
--------	-----------	-----------------------	---

PUT	/cart/:id	Update item quantity	{ quantity }
-----	-----------	----------------------	--------------

Checkout Endpoint

Method	Endpoint	Description	Body
--------	----------	-------------	------

POST	/checkout	Process checkout	{ cartItems, customerInfo }
------	-----------	------------------	-----------------------------

Health Check

Method	Endpoint	Description
--------	----------	-------------

GET	/health	Server status check
-----	---------	---------------------

 Environment Variables

Create a .env file in the backend directory:

```
env
```

```
# MongoDB Connection String
```

```
MONGODB_URI=mongodb://localhost:27017/vibecommerce
```

```
# Server Port
```

```
PORT=5000
```

```
# Environment
```

```
NODE_ENV=development
```

 Usage Guide

>>Adding Products to Cart



>>Click "Add to Cart" on any product

>>View cart by clicking "Cart" in the navigation

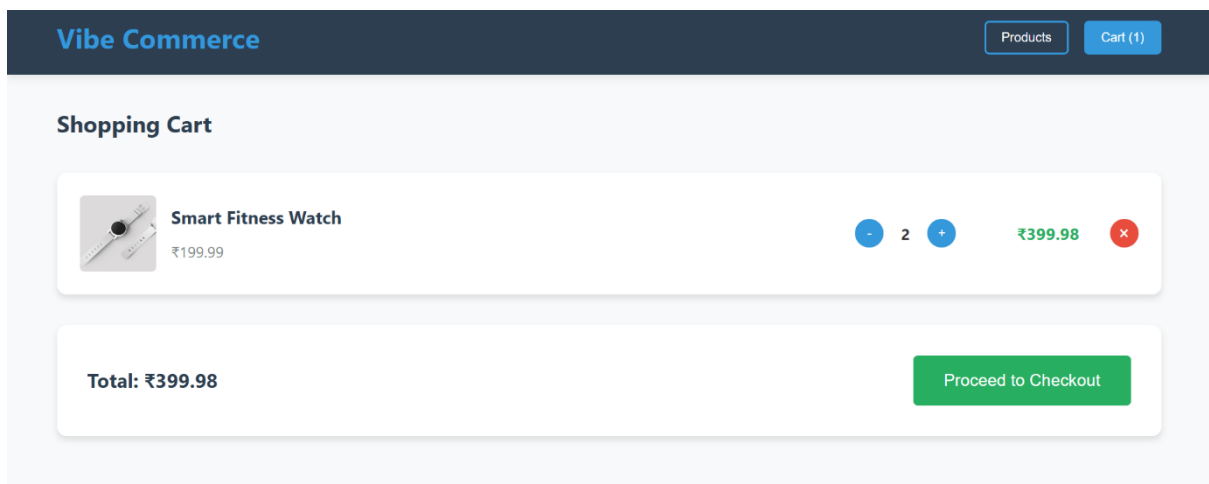


>>Managing Cart Items

>>Update Quantity: Use +/- buttons in cart view

>>Remove Item: Click the × button next to any item

>>View Total: Cart total updates in real-time



>>Checkout Process

>>Click "Proceed to Checkout" from cart view

>>Fill in customer information (name and email required)

Checkout

Order Summary

Smart Fitness Watch × 2₹399.98

Total: ₹399.98

Customer Information

Full Name *

Email *

Address *

City

ZIP Code

Back to Cart

Place Order

>>Submit the form to complete the order

>>View order confirmation receipt

 **Order Confirmed!**

Thank you for your purchase!

Order ID:

ORD-1761709202462-jl3csk48d

Date:

29/10/2025

Time:

09:10:02

Customer Information

Name: CCT-PCEATD-M1

Email: darshanhampali142@gmail.com

Address: Tumkur

Order Items

Smart Fitness Watch

× 2

₹399.98

Total: ₹399.98

A confirmation email has been sent to darshanhampali142@gmail.com

Continue Shopping

Contributing

Fork the project

>Create your feature branch (git checkout -b feature/AmazingFeature)

>Commit your changes (git commit -m 'Add some AmazingFeature')

>Push to the branch (git push origin feature/AmazingFeature)

Open a Pull Request

Authors

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Acknowledgments

>React team for the amazing framework

>MongoDB for the robust database solution

>Unsplash for product images

>Create React App for the quick setup